

VANESSA CEDENO-MIELES, PH.D.

Data Scientist | Machine Learning, Behavioral Analytics & Simulation

Data scientist specializing in predictive modeling, behavioral analytics, and simulation for complex human and organizational systems. Experienced in building Python-based data pipelines and decision-support models for complex organizational systems.

Core Competencies: Workforce analytics • Behavioral data science • Predictive modeling • Experimental design • Machine learning • Data pipelines • Decision-support modeling

Charlottesville, VA | 540-257-4542 | vanessaicm@gmail.com | [LinkedIn](#) | [Website](#) | [Google Scholar](#)

Technical Skills

Machine Learning & Modeling: Predictive analytics, behavioral modeling, agent-based simulation, NLP, statistical modeling.

Data Engineering & Analytics: Python pipelines, ETL workflows, Apache Spark, SQL, large-scale data processing

Visualization & Reporting: Tableau, Power BI, D3.js, interactive dashboards, data storytelling

Programming: Python, R, JavaScript, C++, Java, MATLAB

Research Experience

University of Virginia

Postdoctoral Research Associate

Aug 2024 – Present

Charlottesville, VA

- **Developed a large-scale simulation and machine learning framework modeling 7,500+ stakeholders across the U.S. cement supply chain**, enabling analysis of behavioral adoption dynamics and policy interventions across 51 regional markets.
- **Built Python experimentation pipelines integrating heterogeneous datasets and behavioral modeling** for large-scale scenario analysis.
- **Designed reusable socio-environmental modeling frameworks representing heterogeneous actors, actions, and system dynamics across spatial and temporal scales**, enabling systematic evaluation of policy and organizational coordination mechanisms.
- **Applied statistical modeling and simulation to evaluate complex system transitions**, generating insights on adoption behavior, coordination effects, and strategic interventions in large multi-actor systems.

ESPOL - Escuela Superior Politécnica del Litoral

Associate Professor (2019–2021) — Associate Dean (2022–2024)

Jun 2019 - July 2024

Guayaquil, Ecuador

- **Led research on data-driven modeling and analysis of networked social systems**, designing and deploying FamilySong, a synchronized media platform for distributed families and developing Python-based pipelines to analyze **6 months of behavioral data from 12 distributed households**.
- **Developed data analysis workflows to identify patterns in communication and coordination behavior**, generating insights into how groups sustain long-distance interaction and shared activities.
- **Led data-informed academic operations as Associate Dean**, overseeing 123 faculty and 2,000+ students across 15 academic programs.

- **Designed computational experiments and statistical models to study behavioral dynamics in networked social systems**, combining simulation, data analysis, and machine learning approaches.
- **Processed and analyzed large experimental datasets from networked social science experiments**, developing automated pipelines for data ingestion, feature extraction, and statistical analysis.
- **Identified interaction thresholds influencing collective identity formation in social networks**, demonstrating how communication frequency and group structure influence coordination outcomes.
- **Analyzed behavioral and interaction data to identify patterns in coordination, communication, and group dynamics**, generating insights relevant to organizational behavior and workforce collaboration.

ESPOL – Escuela Superior Politécnica del Litoral
Assistant Professor

May 2010 – Jul 2014
Guayaquil, Ecuador

- Coordinated a multi-year accreditation initiative that resulted in the **first ABET-accredited Computer Science program in Ecuador**, aligning curriculum, and institutional documentation with international accreditation standards.

Publications & Research Impact

12 peer-reviewed publications (7 first-author), with 100+ citations across simulation, machine learning, and network science venues including **CIKM, WSC, ASONAM, and PLOS ONE**.

Full publication list: vcedeno.github.io/papersbyyear.html

Education

Ph.D. in Computer Science - Virginia Tech, Blacksburg, VA
M.S. in Computer Science - Florida State University, Tallahassee, FL
B.S. in Computer Science - ESPOL, Guayaquil, Ecuador

Languages

English (Fluent)
Spanish (Native)
French (Intermediate)